

Start

Business Problem
Key Business Questions
→ Which products generate the most revenue?
→ Which products sell the most units?
→ What time of day does the store perform best?
→ What are the sales trends across products and time intervals?
→ Which products are high-performing and low-performing?
→ What actions can improve sales performance?

Analytics Objectives
OBJECTIVE 1 – REVENUE PERFORMANCE
• Calculate total revenue
• Revenue by product category
• Revenue by product/detail
• Identify highest revenue-generating products
OBJECTIVE 2 – SALES PERFORMANCE
• Calculate total units sold
• Identify best-selling products
• Identify low-performing products
• Compare product performance
OBJECTIVE 3 – TIME ANALYSIS
• Analyse sales by time of day
• Create 30-minute time intervals
• Identify peak sales periods
• Identify slow sales periods
OBJECTIVE 4 – BUSINESS RECOMMENDATIONS
• Recommend actions to increase revenue
• Improve product performance
• Optimise stock levels
• Target slow sales periods
• Support CEO decision-making

Data Source
Excel
→ Daily transactional coffee shop sales data
Key Fields
• Transaction date
• Transaction time
• Product category/type
• Product detail
• Unit price
• Transaction quantity
• Transaction information

Data Ingestion CSV conversion
Excel Dataset
↓
Convert Excel → CSV
↓
Upload CSV
↓
Databricks
↓
Raw/Bronze Data

Load to Databricks

ETL Processing

Data Quality Checkpoint

Bronze Layer
Raw Original imported sales data

Silver Layer
Cleaned and transformed sales data

Gold Layer
Analytics-ready Analysis-ready table
Gold Table Contains:
• Transaction information
• Product information
• Unit price
• Transaction quantity
• Total amount
• Transaction time
• Time bucket
• Product category
• Product detail

SQL Analysis

Key Calculations

KPI Layer

Dashboard & Charts
{ TOTAL REVENUE }
{ TOTAL UNITS SOLD }
{ TOTAL TRANSACTIONS }
{ AVG TRANSACTION VALUE }
{ PEAK SALES PERIOD }
↓
SECOND ROW – TREND ANALYSIS
{ REVENUE TREND }
{ SALES BY TIME INTERVAL }
↓
THIRD ROW – PRODUCT PERFORMANCE
{ REVENUE BY CATEGORY }
{ UNITS SOLD BY PRODUCT }
↓
FOURTH ROW
{ TOP PRODUCTS }
{ LEAST PERFORMING PRODUCTS }
↓
BOTTOM ROW – INSIGHTS
KEY INSIGHTS
• Revenue finding
• Product finding
• Time finding
• Opportunity finding

Key Insights
After completing the analysis, identify:
INSIGHT 1 – REVENUE
Which products/categories contribute the largest share of total revenue?
INSIGHT 2 – PRODUCT PERFORMANCE
Which products are the best sellers?
INSIGHT 3 – LOW PERFORMER
Which products have the lowest sales/revenue?
INSIGHT 4 – TIME
What is the peak sales period?
INSIGHT 5 – SLOW PERIOD
When is customer demand lowest?
INSIGHT 6 – OPPORTUNITY
Which products or time periods provide the biggest

Business Recommendations
RECOMMENDATION 1 – STOCK
Stock more of the best-selling products.
RECOMMENDATION 2 – MARKETING
Run targeted marketing campaigns during slow time slots.
RECOMMENDATION 3 – PRODUCT PROMOTION
Promote underperforming products through:
• Bundles
• Discounts
• Cross-selling
• Product combinations
RECOMMENDATION 4 – PEAK PERIOD
Ensure sufficient stock and staffing during peak sales periods.
RECOMMENDATION 5 – CUSTOMER LOYALTY
Develop loyalty initiatives around peak customer time slots.

Next Steps

CEO Presentation
EXECUTIVE SUMMARY
Present:
→ Overall revenue performance
→ Best-performing products
→ Worst-performing products
→ Peak sales periods
→ Major opportunities
→ Recommended actions

ANALYSIS OBJECTIVES
Show how the analysis answered:
• Which products generate the most revenue?
• Which products sell the most?
• When does the store perform best?
• Where are the performance gaps?
• How can sales improve?

SHORT TERM
• Optimise stock
• Promote slow-selling products
• Target slow periods

MEDIUM TERM
• Introduce targeted campaigns
• Improve product mix
• Optimise staffing

LONG TERM
• Automate daily sales reporting
• Track performance across multiple locations
• Implement loyalty programmes

Final Deliverables

MIRO
Data Flow + Architecture + Analysis Plan

DATABRICKS
Cleaned/processed dataset + SQL analysis

EXCEL / VISUALISATION TOOL
Pivot Tables + Charts + Dashboard

POWERPOINT
CEO Presentation

SQL
Complete SQL Analysis File

GITHUB
Final project repository containing:

• Miro planning
• Dataset/processed dataset
• SQL scripts
• Dashboard

• PowerPoint presentation

Scan the QR code or go to and enter the code